JMMMU: A Japanese Massive Multi-discipline Multimodal Understanding Benchmark for Culture-aware Evaluation

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Abstract

Multimodal benchmarks for evaluating Large Multimodal Models (LMMs) are predominantly English-centric, and existing Japanese benchmarks remain small in scale and focus on common-sense tasks, failing to assess expert-level or culture-specific knowledge. We introduce JMMMU (Japanese Massive Multi-discipline Multimodal Understanding), the first large-scale, expert-level Japanese multimodal benchmark, comprising 1,320 questions and 1,118 images across 28 subjects. JMMMU disentangles language ability from cultural knowledge by splitting into two complementary subsets: a *Culture-Agnostic* (CA) subset of 720 questions translated and localized from MMMU, and a *Culture-Specific* (CS) subset of 600 newly authored questions covering Japanese Art, Japanese Heritage, Japanese History, and World History as taught in Japan. The CA subset enables direct English–Japanese performance comparisons, while the CS subset probes whether models genuinely understand Japanese cultural knowledge. We evaluate 16 open-source and 3 proprietary LMMs. Even the best proprietary model (GPT-4o) achieves only 58.6% overall, while the best open-source model (LLaVA-OV-7B) reaches 40.5%. On the CA subset, we observe an average accuracy drop of 4.3% when switching from English to Japanese, even with identical content, revealing a systematic language bias. On the CS subset, large disparities emerge even among models with similar CA scores—for example, GPT-4o outperforms Claude 3.5 Sonnet by 15.7% on CS despite comparable CA performance. These findings demonstrate that strong Japanese language handling does not imply cultural competence, and that both dimensions must be measured to fairly assess and advance multilingual multimodal systems. We release the benchmark to facilitate future research.

1 Introduction

Large Multimodal Models (LMMs) have made rapid progress across a wide variety of vision-language tasks (Liu et al., 2023; Li et al., 2024; Chen et al., 2024; OpenAI, 2024; Reid et al., 2024). However, the benchmark landscape used to track this progress remains heavily English-centric. While rigorous benchmarks such as MMMU (Yue et al., 2024a) test expert-level, college-grade reasoning across multiple disciplines, their design reflects a predominantly English-speaking cultural context. Models that appear strong on such benchmarks may nonetheless fail when asked to reason about knowledge grounded in other linguistic or cultural contexts.

Japanese is an important case study. It ranks among the most-studied languages after English, yet the current landscape of Japanese multimodal benchmarks is limited. Existing resources such as JA-VG-VQA-500 (SakanaAI, 2024b), JA-VLM-Bench-In-the-Wild (SakanaAI, 2024c), JA-Multi-Image-VQA (SakanaAI, 2024a), LLaVA-Bench-in-the-Wild (JA) (Turing, 2024), and Heron-Bench (Inoue et al., 2024b) are all small in scale (at most 102 questions) and focus on common-sense or everyday visual tasks rather than expert-level reasoning (see Table 1). None of them provide the diagnostic power needed to evaluate whether models have internalized Japanese cultural knowledge at a college or expert level.

This gap creates a serious measurement problem. A model might score well on translated English questions in Japanese (i.e., it has strong Japanese *language* ability) but perform poorly on questions that genuinely require knowledge of Japanese culture, history, or art. Conversely, a model could exhibit the reverse pattern. Without a benchmark that separates these two axes, we cannot accurately diagnose model capabilities or guide the development of truly culturally-aware multilingual LMMs.

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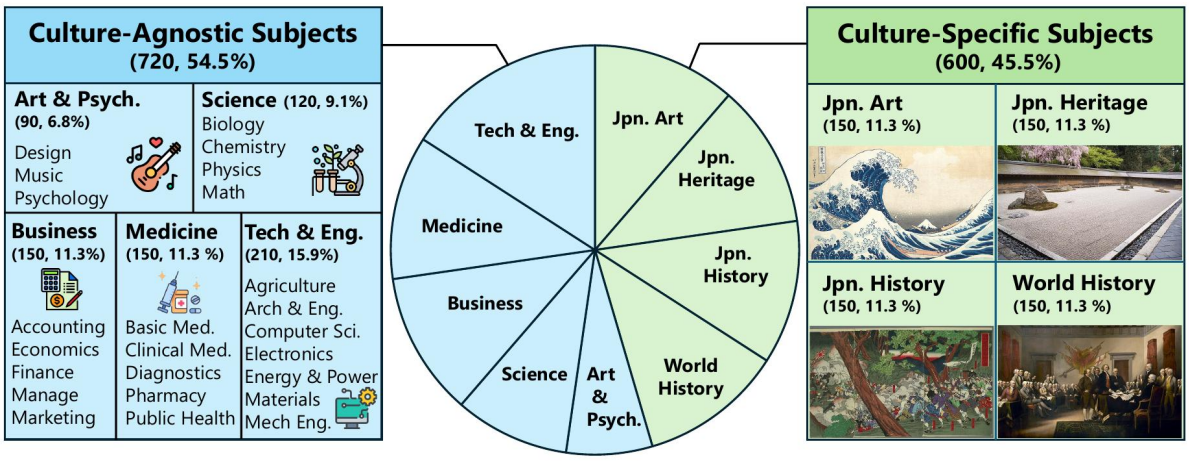

Figure 1: **Overview of the JMMMU dataset.** JMMMU comprises 1,320 questions with 1,118 images across 28 subjects, split into a culture-agnostic (CA) subset translated from MMMU and a culture-specific (CS) subset with four new Japanese-focused subjects. The benchmark supports both multiple-choice and open-ended question formats.

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To address this, we introduce **JMMMU** (Japanese Massive Multi-discipline Multimodal Understanding), a large-scale benchmark designed specifically for culture-aware evaluation of Japanese LMMs. JMMMU contains 1,320 questions with 1,118 images and is organized around a clear conceptual split:

- **Culture-Agnostic (CA) subset** (720 Q, 24 subjects): Questions and images from MMMU are carefully translated and localized into Japanese. This subset enables clean English–Japanese comparisons on identical content, isolating the impact of language on model performance.
- **Culture-Specific (CS) subset** (600 Q, 4 subjects): Entirely new questions covering Japanese Art, Japanese Heritage, Japanese History, and World History as taught in Japan. This subset requires genuine cultural knowledge that cannot be answered through language facility alone.

Figure 1 provides an overview of the dataset.

Through comprehensive evaluation of 19 LMMs (16 open-source and 3 proprietary), JMMMU reveals several important findings: (1) Top models leave substantial room for improvement—GPT-4o reaches only 58.6% and the best open-source model LLaVA-OV-7B reaches 40.5%. (2) There is a systematic language performance gap: switching from English to Japanese on identical CA content degrades average accuracy by 4.3%, even for state-of-the-art models. (3) Cultural knowledge is poorly correlated with general language ability: models cluster differently on the CA and CS subsets, confirming that they measure distinct capabilities. (4) The Heritage subject is the hardest, with the best open-source model scoring 40.0% versus GPT-4o's 70.7%.

Contributions. We summarize our contributions as:

- We introduce JMMMU, the first large-scale (1,320 Q; 1,118 images), expert-level Japanese multimodal benchmark with both culture-agnostic and culture-specific evaluation.
- We propose a principled construction pipeline that includes professional image localization and new culturally-grounded content curated by 19 expert native annotators.
- We provide a comprehensive evaluation of 19 diverse LMMs, disentangling language handling from cultural knowledge and revealing systematic capability gaps.
- We conduct detailed analyses of translation effects, error types, and score correlations to guide the development of culturally-aware multilingual multimodal systems.

2 Related Work

Expert-Level Multimodal Benchmarks. MMMU (Yue et al., 2024a) is the most prominent expert-level multimodal benchmark, featuring 11.5K college-level questions across 30 subjects in English. MMMU-Pro (Yue et al., 2024c) extends it with harder settings. CMMMU (Zhang et al., 2024) provides a Chinese adaptation following the same design principles. MMBench (Liu et al., 2024b) evaluates models on a broader set of multimodal

skills. JMMMU complements these resources by providing the first such benchmark for Japanese, including culture-specific content that cannot be obtained through translation alone.

Multilingual and Cross-Cultural Benchmarks. Several efforts have extended multimodal evaluation beyond English. xGQA (Pfeiffer et al., 2021) and MaXM (Changpinyo et al., 2022) extend English VQA datasets to multiple languages via translation, but rely on Western-centric images. MaRVL (Liu et al., 2021) introduces culturally-grounded images for diverse languages through native-speaker annotation. CVQA (Romero et al., 2024) covers 30 countries and 31 languages with culturally-driven content. MTVQA (Tang et al., 2024) focuses on multilingual text-centric VQA. Pangea (Yue et al., 2024b) provides an open multilingual multimodal LLM covering 39 languages along with a multilingual benchmark. MIA-Bench (Qian et al., 2024) evaluates instruction following in multimodal LLMs. While these works highlight the importance of multilingual evaluation, none provides expert-level assessment in Japanese or systematically disentangles language ability from cultural knowledge.

Japanese Multimodal Benchmarks. Existing Japanese multimodal benchmarks are small and limited to common-sense tasks. JA-VG-VQA (Shimizu et al., 2018) and its 500-sample variant JA-VG-VQA-500 (SakanaAI, 2024b) provide bilingual VQA data. LLaVA-Bench-in-the-Wild (JA) (Turing, 2024) contains only 60 questions. JA-VLM-Bench-In-the-Wild (SakanaAI, 2024c) and JA-Multi-Image-VQA (SakanaAI, 2024a) each cover fewer than 60 questions with a cultural focus. Heron-Bench (Inoue et al., 2024b) offers 102 Japanese-context image-question pairs for evaluating Japanese VLMs. Table 1 summarizes these benchmarks. JMMMU is the first to provide expert-level evaluation at a large scale for Japanese, combining culture-agnostic and culture-specific content.

Large Multimodal Models. Recent open-source LMMs include LLaVA-1.5 (Liu et al., 2024a), LLaVA-OneVision (LLaVA-OV) (Li et al., 2024), InternVL2 (Chen et al., 2024), CogVLM2 (Hong et al., 2024), Idefics2 (Laurençon et al., 2024b), Idefics3 (Laurençon et al., 2024a), Mantis (Jiang et al., 2024), and xGen-MM (Xue et al., 2024). Japanese-specific LMMs include LLaVA-CALM2 (Inagaki, 2024) and EvoVLM-JP-v2 (Inoue et al., 2024a), the latter developed using evolutionary model merging (Akiba et al., 2024). Pangea (Yue et al., 2024b) is a multilingual LMM supporting 39 languages. Proprietary models include GPT-4o (OpenAI, 2024), Gemini 1.5 Pro (DeepMind, 2024; Reid et al., 2024), and Claude 3.5 Sonnet (Anthropic, 2024). We evaluate all of these models on JMMMU to provide a comprehensive picture of the state of the art.

Table 1: **Overview of Japanese LMM benchmarks.** JMMMU is the first benchmark that evaluates expert-level skills and is the largest among culture-aware benchmarks.

Benchmark	Culture	Level	Questions	Images
JA-VG-VQA-500 (SakanaAI, 2024b)	✗	Common sense	500	500
LLaVA-Bench-in-the-wild (Turing, 2024)	✗	Common sense	60	24
JA-Multi-Image-VQA (SakanaAI, 2024a)	✓	Common sense	55	39
JA-VLM-Bench-in-the-wild (SakanaAI, 2024c)	✓	Common sense	50	42
Heron Bench (Inoue et al., 2024b)	✓	Common sense	102	21
JMMMU (Ours)	✓	Expert	1,320	1,118

3 JMMMU Benchmark

3.1 Design Principles

JMMMU is built around two core design principles. First, it should enable *language-culture disentanglement*: by including both translated (culture-agnostic) and natively authored (culture-specific) content, the benchmark allows researchers to determine whether a model's Japanese performance reflects language competence, cultural knowledge, or both. Second, it should provide *expert-level difficulty*: questions are drawn from or inspired by college-level curricula, requiring domain-specific reasoning beyond simple visual recognition or common-sense understanding.

The dataset contains 1,320 questions with 1,118 images spanning 28 subjects organized under five broad categories: Culture-Specific (CS), Art & Psychology, Business, Science, Health & Medicine, and Tech & Engineering. Questions come in multiple-choice (MC) format with four answer options and open-ended format requiring a short phrase.

3.2 Culture-Agnostic (CA) Subset

The CA subset consists of 720 questions across 24 subjects translated and localized from the MMMU validation set. To construct it, we first identify the 24 subjects in MMMU that are culture-neutral—those whose questions do not rely on English-language or Western cultural context. Six subjects

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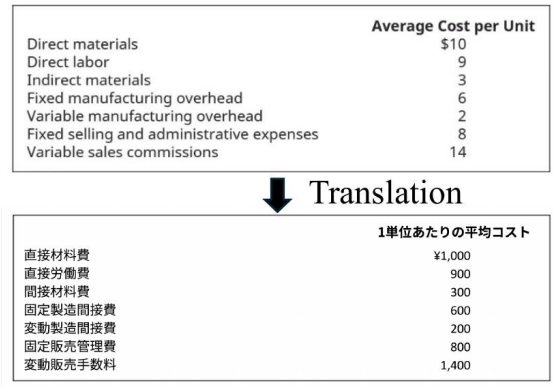

Figure 2: **Example of the image translation process.** English text embedded in images (left) is manually overwritten with Japanese (right). This ensures that the CA subset is fully localized and avoids confounding image-language with question-language.

in MMMU (Art, Art Theory, Geography, History, Literature, Sociology) were excluded as culturally dependent, leaving 24 suitable subjects.

Text Localization. All question text, answer options, and explanatory text are first automatically translated using GPT-4o (OpenAI, 2024), then manually refined by subject-expert native Japanese annotators for naturalness, domain terminology conventions, and appropriate style. Monetary values (e.g., $1 → ¥100) are normalized to fit Japanese norms without altering the mathematical difficulty of questions.

Image Localization. A key contribution of our CA construction is the systematic localization of image-embedded text. When MMMU images contain English text (e.g., chart labels, diagram captions, mathematical notation, table headers), we manually overwrite that text with Japanese using image editing tools. Figure 2 illustrates this process. This step is necessary to avoid confounding language proficiency with the visual reasoning being tested—a model should fail a Japanese question because of insufficient Japanese knowledge, not because image-embedded text remains in English.

3.3 Culture-Specific (CS) Subset

The CS subset consists of 600 newly authored questions across four subjects not present in MMMU: **Japanese Art** (150 Q), **Japanese Heritage** (150 Q), **Japanese History** (150 Q), and **World History** (150 Q, aligned to Japanese high-school and university curricula). These subjects require knowledge that is genuinely rooted in Japanese culture and cannot be correctly answered through language facility alone.

Images for the CS subset are sourced from Wikimedia Commons under redistributable licenses, covering paintings, artifacts, historical documents, cultural sites, and historical events. Question authoring followed guidelines requiring: (1) concise question text that focuses on a single concept, (2) answer options that are plausible distractors balanced so that visual grounding is necessary to discriminate, and (3) alignment with standard university-level knowledge in Japan.

3.4 Annotation and Quality Control

The benchmark was constructed with the involvement of 19 university students including the authors, all native Japanese speakers with subject-area expertise. The annotation pipeline included: (1) initial automatic translation (GPT-4o) followed by human expert refinement, (2) independent review of translated items for domain terminology accuracy and cultural correctness, (3) authoring of CS questions with feedback loops ensuring option plausibility and visual dependency, and (4) a final round of review for fluency and overall quality.

3.5 Evaluation Protocol

All evaluations use Japanese-language prompts. For multiple-choice questions, the model is asked to output the option letter (A/B/C/D). For open-ended questions, the model outputs a short phrase. A rule-based parser extracts the answer from model outputs. If parsing fails (e.g., due to refusal or non-standard formatting), the question is assigned a random choice to avoid rewarding abstention. This evaluation protocol is consistent across all models to ensure fair comparison.

4 Experiments

4.1 Experimental Setup

We evaluate 19 LMMs: 16 open-source models and 3 proprietary models. Open-source models include LLaVA-OV-0.5B and LLaVA-OV-7B (Li et al., 2024), InternVL2-2B and InternVL2-8B (Chen et al., 2024), xGen-MM (Xue et al., 2024), Phi-3v and Phi-3.5v (Abdin et al., 2024), LLaVA-1.6-13B and LLaVA-1.6-34B (Liu et al., 2024a), Idefics2-8B (Laurençon et al., 2024b), Idefics3-8B (Laurençon et al., 2024a), Mantis-8B (Jiang et al., 2024), CogVLM2-19B (Hong et al., 2024),

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Pangea-7B (Yue et al., 2024b), and two Japanese-tuned models: LLaVA-CALM2 (Inagaki, 2024) and EvoVLM-JP-v2 (Inoue et al., 2024a). Proprietary models are GPT-4o (OpenAI, 2024), Gemini 1.5 Pro (DeepMind, 2024), and Claude 3.5 Sonnet (Anthropic, 2024). We also report a text-only GPT-4o baseline (no image input) to assess the degree to which questions are visually grounded. All models receive questions in Japanese using a standardized prompt format.

4.2 Main Results

Table 2 presents the overall results. We make the following key observations.

Overall difficulty. JMMMU is challenging across the board. The best proprietary model, GPT-4o, achieves 58.6% overall, and the best open-source model, LLaVA-OV-7B, achieves 40.5%—both substantially above the random baseline of 24.8% but leaving significant headroom for improvement.

Proprietary vs. open-source gap. The proprietary models substantially outperform open-source models overall: GPT-4o (58.6%) > Gemini 1.5 Pro (51.5%) > Claude 3.5 Sonnet (50.8%), while the top open-source model reaches only 40.5%. This gap is especially pronounced on the CS subset, where GPT-4o (66.7%) leads Claude 3.5 Sonnet (51.0%) by 15.7% absolute, despite similar CA scores (GPT-4o: 51.8%, Claude: 50.6%).

Culture-Agnostic vs. Culture-Specific. Comparing CA and CS scores reveals interesting patterns. Gemini 1.5 Pro shows a large CS–CA gap (60.3% CS vs. 44.2% CA), while Claude 3.5 Sonnet achieves nearly equal scores (51.0% CS, 50.6% CA). Among open-source models, Japanese-tuned LMMs (LLaVA-CALM2: CS 41.5%, CA 29.4%; EvoVLM-JP-v2: CS 45.2%, CA 32.2%) are competitive on CS relative to their CA scores, suggesting that Japanese-specific training helps cultural knowledge.

Subject-level variation. World History is the highest-scoring CS subject for most models (LLaVA-OV-7B: 68.0%; GPT-4o: 76.7%), as it overlaps with general world knowledge. Japanese Heritage is consistently hardest (LLaVA-OV-7B: 30.7%; GPT-4o: 70.7%)—a 40-point gap for the best models. Science and Tech & Engineering are relatively challenging for most open-source models.

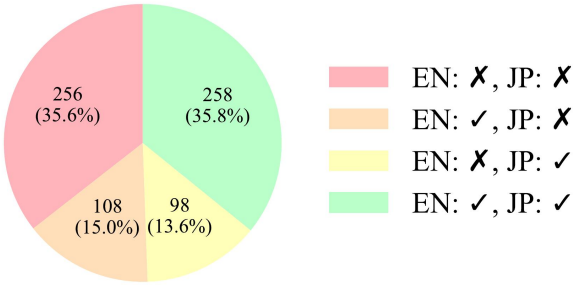

Figure 3: Breakdown of GPT-4o performance on the CA subset. A considerable fraction of questions are answered correctly only in English or only in Japanese (yellow + orange), illustrating the language performance gap even when the underlying knowledge is language-neutral.

Text-only baseline. GPT-4o without image input achieves 38.1% overall, well below its 58.6% with images, confirming that JMMMU questions generally require visual grounding. The gap is especially large on CS (text-only: 35.5% vs. 66.7%), showing that cultural knowledge must be visually reasoned, not merely recalled from text.

5 Analysis

5.1 Language vs. Culture Disentanglement

A central property of JMMMU is the ability to separate language ability from cultural knowledge using the CA subset. Figure 3 illustrates GPT-4o's behavior on the CA subset: a substantial fraction of questions are answered correctly only in English or only in Japanese (yellow and orange segments), not in both—demonstrating that even with identical content, language matters.

5.2 Effect of Translation

To understand how text and image localization each contribute to the performance gap, we conduct an ablation study on the CA subset using a representative set of models. We compare three conditions: (1) $I_{en}T_{en}$: both image and text in English (original MMMU), (2) $I_{en}T_{jp}$: translated text but original English images, and (3) $I_{jp}T_{jp}$: both image and text translated to Japanese (the full JMMMU CA condition). Table 3 shows the results.

For most models, translating the text alone ($I_{en}T_{jp}$) already degrades performance compared to the English baseline: e.g., LLaVA-OV-7B drops from 45.0 to 38.3 (-6.7%) and InternVL2-8B from 43.9 to 38.3 (-5.6%). Further translating the images ($I_{jp}T_{jp}$) leads to additional drops: LLaVA-

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Table 2: **Overall results.** CA (EN) shows the result on culture agnostic subset in English. The rest of the results are average and individual subjects' scores on JMMMU. †denotes Japanese LMMs. The best-performing model among open source and proprietary models are **in bold**. Overall, the performance is up to 40.5% for open-source, and 58.6% for proprietary models, leaving great room for improvement.

Models	Overall (1,320)	CS (600)	CA (720)	CA (EN) (720)	Jpn. Art (150)	Jpn. Heritage (150)	Jpn. History (150)	World History (150)	Art & Psych. (90)	Business (150)	Science (120)	Health & Medicine (150)	Tech & Eng. (210)
Random	24.8	25.0	24.6	24.6	25.0	25.0	25.0	25.0	25.4	25.0	22.8	25.6	24.3
Open Source													
LLaVA-OV-0.5B	26.0	23.3	28.2	29.4	22.7	22.7	24.0	24.0	26.7	27.3	24.2	30.7	30.0
InternVL2-2B	28.3	29.2	27.6	31.9	31.3	22.7	30.7	32.0	30.0	30.0	30.8	25.3	24.8
xGen-MM	28.6	28.2	28.9	35.7	30.0	20.7	22.7	39.3	32.2	21.3	22.5	36.7	31.0
Phi-3v	29.5	26.5	31.9	37.6	31.3	18.7	29.3	26.7	26.7	28.7	25.8	37.3	36.2
LLaVA-1.6-13B	31.1	33.7	29.0	29.9	32.0	24.0	32.0	46.7	25.6	28.7	30.0	34.0	26.7
Idefics2-8B	31.9	37.0	27.6	35.1	40.7	24.0	30.0	53.3	32.2	22.7	22.5	32.0	29.0
Phi-3.5v	32.4	34.3	30.8	39.2	37.3	27.3	35.3	37.3	27.8	31.3	30.0	36.7	28.1
†LLaVA CALM2	34.9	41.5	29.4	29.9	42.7	36.7	40.0	46.7	27.8	26.0	26.7	34.0	31.0
Mantis 8B	35.5	39.5	32.2	36.0	42.0	30.0	35.3	50.7	37.8	28.0	31.7	37.3	29.5
CogVLM2-19B	36.1	39.7	33.1	36.8	39.3	24.0	36.0	59.3	28.9	32.7	30.8	30.0	38.6
Idefics3-8B	37.3	42.8	32.8	36.9	43.3	24.7	**42.0**	61.3	34.4	28.0	26.7	38.0	35.2
†EvoVLM JP v2	38.1	45.2	32.2	33.9	44.0	**40.0**	**42.0**	54.7	32.2	28.7	28.3	**38.7**	32.4
InternVL2-8B	38.3	42.5	34.7	43.3	41.3	38.0	35.3	55.3	40.0	36.0	**34.2**	34.0	32.4
Pangea-7B	39.7	**47.0**	33.6	35.9	**46.0**	35.3	40.0	66.7	40.0	30.0	31.7	43.3	27.6
LLaVA-1.6-34B	39.8	43.2	37.1	45.7	42.0	36.0	40.7	54.0	**42.2**	**41.3**	25.0	36.7	39.0
LLaVA-OV-7B	**40.5**	43.0	**38.5**	45.1	36.0	30.7	37.3	**68.0**	41.1	36.7	31.7	**38.7**	**42.4**
Proprietary													
Claude 3.5 Sonnet	50.8	51.0	50.6	52.1	39.3	46.7	54.7	63.3	**53.3**	**56.7**	**51.7**	55.3	41.0
Gemini 1.5 Pro	51.5	60.3	44.2	51.1	54.7	55.3	55.3	76.0	51.1	44.0	44.2	48.0	38.6
GPT-4o	**58.6**	**66.7**	**51.8**	52.1	**60.7**	**70.7**	**58.7**	**76.7**	**53.3**	55.3	45.8	**61.3**	**45.2**
Text Only													
GPT-4o text	38.1	35.5	40.3	44.9	32.7	32.0	35.3	42.0	38.9	36.0	41.7	45.3	39.5

Table 3: **The effect of translation.** Each column shows the model performance when image (I) and text (T) are in Japanese (jp) or in English (en). Δ_i shows the difference from $I_{en}T_{en}$.

Model	$I_{en}T_{en}$	$I_{en}T_{jp}(\Delta_1)$	$I_{jp}T_{jp}(\Delta_2)$
LLaVA-1.6-13B	26.4	31.9 (+5.5)	29.2 (+2.8)
Phi-3.5v	39.2	33.6 (-5.6)	31.1 (-8.1)
LLaVA-CALM2	29.4	28.3 (-1.1)	31.4 (+2.0)
CogVLM2-19B	32.8	31.9 (-0.9)	34.4 (+1.6)
EvoVLM JP v2	30.0	30.8 (+0.8)	28.6 (-1.4)
InternVL2-8B	43.9	38.3 (-5.6)	37.2 (-6.7)
LLaVA-1.6-34B	43.6	40.8 (-2.8)	38.9 (-4.7)
LLaVA-OV-7B	45.0	38.3 (-6.7)	35.6 (-9.4)

OV-7B falls to 35.6 (−9.4% from the English baseline). These results indicate that both text-level language ability and the ability to read Japanese text in images are bottlenecks for current models. Notably, Japanese-specialized models such as LLaVA-CALM2 and EvoVLM-JP-v2 are more robust to these changes, confirming the value of targeted Japanese training.

5.3 Score Correlation Between Subsets

Figure 4 plots each model's CA score against its CS score. Although a positive correlation exists (stronger models tend to perform better on both), the relationship is far from deterministic. This supports the core claim of JMMMU: cultural knowledge and general Japanese language ability are distinct capabilities that require separate measurement. Notably, Gemini 1.5 Pro lies significantly above the trend line (high CS relative to CA), while certain models cluster together on CA but diverge widely on CS.

5.4 Error Analysis on Culture-Specific Questions

To understand where models fail on the CS subset, we conduct a manual error analysis on GPT-4o's incorrect answers, classifying them into four types:

- **Lack of Knowledge**: The model fails to identify the correct answer because it lacks knowledge of the Japanese cultural item depicted (e.g., misidentifying a historical artifact).
- **Textual Misunderstanding**: The model misinterprets the Japanese question or option text, lead-

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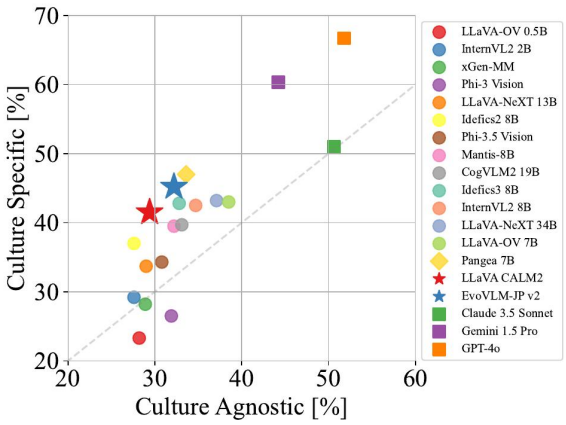

Figure 4: **Score correlation between the CA and CS subsets.** While there is a positive trend, significant scatter exists—particularly among top models—confirming that CA and CS measure distinct competencies. Models can score similarly on CA but differ substantially on CS.

ing to the wrong answer despite having the relevant knowledge.

- **Image Recognition Error**: The model incorrectly identifies or describes the visual content, such as confusing similar artworks or cultural objects.
- **Answer Rejection**: The model refuses to answer or produces a non-answer response that the parser cannot resolve.

Figure 5 shows the distribution of error types across the four CS subjects. Lack of Knowledge is the dominant error for Japanese Heritage and Japanese History, where detailed cultural knowledge is required. Image Recognition Errors are relatively more frequent in Japanese Art, where many artworks appear visually similar. Textual Misunderstanding and Answer Rejection are less common but non-negligible, pointing to room for improvement in Japanese language handling even for GPT-4o.

5.5 Discussion

The results from JMMMU lead to several actionable conclusions for the multilingual LMM community.

Language ≠ Culture. The systematic gap between CA and CS scores, and the weak correlation between English-to-Japanese degradation and CS performance, confirms that Japanese language ability and Japanese cultural knowledge are distinct and must be evaluated separately. Improving a

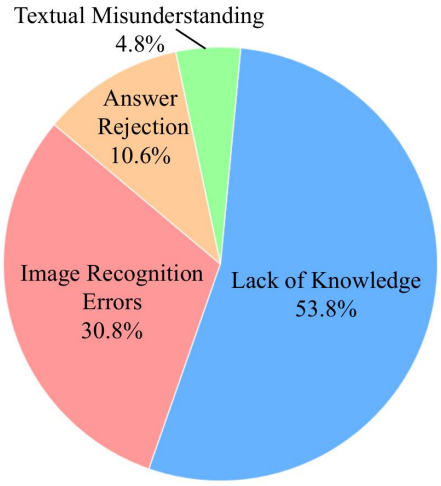

Figure 5: **Error distribution over culture-specific subjects.** Error types—Lack of Knowledge, Textual Misunderstanding, Image Recognition Error, and Answer Rejection—vary across subjects. Lack of Knowledge is dominant in Japanese Heritage and History, while Image Recognition Errors are more prominent in Japanese Art.

model's Japanese fluency does not automatically yield cultural competence.

Japanese-tuned models reduce the language gap but not the culture gap. Japanese-specialized models (LLaVA-CALM2, EvoVLM-JP-v2) reduce the English-to-Japanese performance gap on the CA subset relative to general-purpose models. However, even these models perform far below proprietary baselines on the CS subset, showing that dedicated cultural training data remains scarce and needed.

Image localization matters. The translation ablation (Table 3) shows that image localization contributes meaningfully to performance degradation in current models. This suggests that models trained predominantly on English-text images are disadvantaged by Japanese in-image text, and that future multilingual training datasets should include more diverse in-image language coverage.

Benchmark limitations. JMMMU is limited to 28 subjects and does not cover all aspects of Japanese culture or all expert domains. The CS subjects reflect a university curriculum perspective and may not capture vernacular or contemporary cultural knowledge. No inter-annotator agreement was formally computed. As with any benchmark, there is a risk of data contamination for propri-

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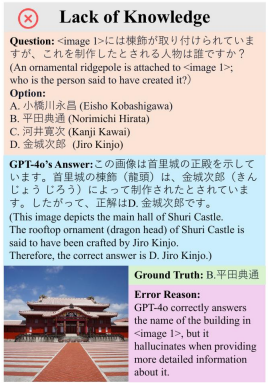
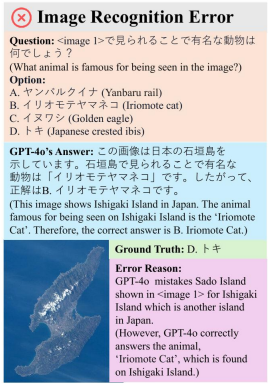
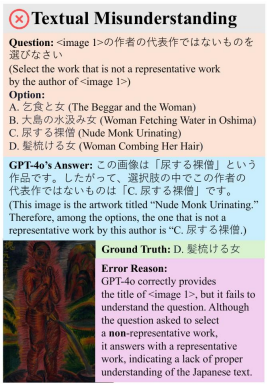


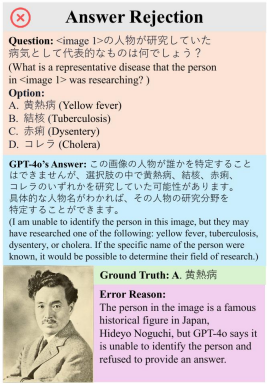
(a) Lack of Knowledge



(b) Image Recognition Error



(c) Textual Misunderstanding



(d) Answer Rejection

Figure 6: **Examples of each error type on culture-specific questions.** (a) Lack of Knowledge: the model guesses incorrectly due to missing cultural knowledge. (b) Image Recognition Error: the model misidentifies a visual artifact. (c) Textual Misunderstanding: the model answers a different question due to incorrect parsing of Japanese text. (d) Answer Rejection: the model declines to answer or generates an unparsable response.

etary models. Nevertheless, the scale and design of JMMMU represent a substantial step forward for Japanese multimodal evaluation, and we release it to enable reproducible future work.

6 Conclusion

We have presented JMMMU, the first large-scale, expert-level Japanese multimodal benchmark, containing 1,320 questions and 1,118 images across 28 subjects. JMMMU uniquely disentangles language ability from cultural knowledge by pairing a translated culture-agnostic subset with a natively authored culture-specific subset. Comprehensive evaluation of 19 LMMs reveals that current models are far from solving the benchmark, that language ability and cultural knowledge are meaningfully distinct, and that the Japanese Heritage domain poses the greatest challenge. We hope JMMMU will spur the development of LMMs that are not merely multilingual but genuinely multicultural.

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